

AI / BACKEND ENGINEER

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PROFESSIONAL SUMMARY

Backend-focused AI / software engineer specializing in Python and FastAPI — a year building a production GPT-4 translation system for EdTech, full-stack delivery of a university flagship banking web/app, and earlier Docker/production work. Builds LLM infrastructure in the open: cost-aware model routing, multi-agent workflows, evaluation and guardrails. MSc Data Science student in Berlin; competitive security/CTF background (1st place, IIUM CTF).

PROFESSIONAL EXPERIENCE

Full-Stack Developer — XtraLesson (EdTech)

Mar 2025 – Feb 2026 · Qatar · Remote

Python · FastAPI · OpenAI GPT-4 · PostgreSQL · async pipelines · CI/CD

- Built an async, fault-tolerant OpenAI GPT-4 translation system for a 50-course library, with retry logic that reprocessed only failed blocks — cutting API calls by roughly a third and turning days of processing into hours.
- Implemented a real-time parsing/job tracker (Start → Process → Complete) that gave non-technical teams live visibility of translation progress and removed manual status requests.
- Refactored translation into user-triggered actions and customized frontend branding to XtraLesson's identity, improving flexibility and engagement.

App & Web Developer — IIUM Flagship 18: Sustainable Banking

2023 · ~6 months · International Islamic University Malaysia · Kuala Lumpur

Full-stack web & mobile · deployment · domain migration

- Sole developer for the university's Flagship 18 sustainable-banking initiative — built the entire website and mobile application end to end.
- Migrated the platform to its own dedicated domain, handling deployment and the full transition.

Junior Software Engineer — WenakLabs

Mar 2020 – Feb 2021 · N'Djamena, Chad · On-site

Docker · production systems · release & infrastructure

- Developed and maintained applications supporting core production systems.
- Managed containerization with Docker to streamline deployment and scalability.

EDUCATION

MSc Data Science & Business Intelligence

GISMA Business School, Berlin, Germany · GPA 3.5

Oct 2025 – present

BSc Information Technology (Hons.) — Business Intelligence & Information Science

International Islamic University Malaysia (IIUM) · GPA 4.0 / CGPA 3.2

2021 – 2025

PROJECTS (BUILDING IN THE OPEN)

RouteIQ — LLM Cost Router & Observability Gateway **SHIPPED** · Python, FastAPI, Redis, Prometheus, Grafana · github.com/yaya-ali/routeiq



- Shipped an API gateway that routes each LLM request to the cheapest capable model — measured **10.2% cost saving with 0% quality drop** on an eval set, enforced by an eval-gated CI pipeline that blocks quality regressions.
- Per-team daily budgets via atomic date-keyed Redis counters (HTTP 429 on exhaustion, UTC-midnight reset); Prometheus metrics with auto-provisioned Grafana dashboard; chat and quiz demo UIs as consumer apps.

Multi-Agent Content Pipeline **SHIPPED** · Next.js, TypeScript · github.com/yaya-ali/multi-agent-content-pipeline

- Five-stage agent flow — research, write, fact-check, polish, grade — that turns a topic into a reviewed article in one run, with failed fact-checks routed back to the writer automatically.

Medical GPT Assistant **SHIPPED** · Python, Docker, H2OGPT, TypeScript · github.com/yaya-ali/medicalAI

- Self-hosted medical LLM assistant (Dockerized H2OGPT) for patient triage at a clinic — preprocessing pipeline cut response time ~40%; adopted into the daily workflow by 3 doctors.

Terminal Dashboard **SHIPPED** · Python, Textual (TUI) · github.com/yaya-ali/terminal-dashboard

- Single-screen terminal dashboard for developers juggling multiple projects — git repos, sessions, ports and recent activity in one live TUI.

more projects — github.com/yaya-ali · yaya-ali.github.io

TECHNICAL SKILLS

Languages Python, SQL, TypeScript, JavaScript, PHP

Backend & APIs FastAPI, Pydantic, REST API design, microservices, asyncio & async job queues, Node.js / Express

AI / LLM RAG, embeddings & vector search (pgvector), prompt engineering, multi-agent systems (LangChain, LangGraph, LlamaIndex), eval harnesses, guardrails / red-teaming, model routing, OpenAI & Anthropic APIs, PyTorch, scikit-learn

Databases PostgreSQL, Supabase (pgvector), Redis, MongoDB — schema design, query optimization, indexing

Cloud & DevOps AWS, Azure, Google Cloud, Docker, Kubernetes, GitHub Actions (CI/CD), Terraform, Prometheus, Grafana

Frontend React, Next.js, Tailwind CSS

Practices Agile / Scrum, code review, automated testing (pytest), Git, Postman, Linux

CERTIFICATIONS & LANGUAGES

Certifications: Cisco CCNA · AWS Cloud Practitioner · FastAPI & REST APIs (Udemy) · Intro to Cybersecurity (Cisco) · 1st Place — IIUM CTF (2022) · Top 20% — PicoCTF Global (2023)

Languages: Arabic (native) · English (fluent) · French (intermediate) · German (basic, improving)